Quiz 4

Graded

Student

Clarisse Cheung

Total Points

0 / 4 pts

Question 1

Question 1

0 / 1 pt

- **– 0 pts** Full Credit
- **– 0.1 pts** Majority Credit
- **– 0.5 pts** Half Credit
- ✔ **– 1 pt** No Credit

1. This doesn't prove $-1 \notin QR_p$.
2. I recommend using Euler's Criterion.

Question 2

Question 2

0 / 1 pt

- **– 0 pts** Full Credit
- **– 0.1 pts** Majority Credit
- **– 0.5 pts** Half Credit
- ✔ **– 1 pt** No Credit

3. I recommend proving this statment by using Euler's Criterion. Note that $(-y)^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{2}} \cdot y^{\frac{p-1}{2}}$. Then use part (a) to complete the proof.

Question 3

Question 3

0 / 2 pts

- **– 0 pts** Full Credit
- **– 0.2 pts** Majority Credit
- **– 1 pt** Half Credit
- ✔ **– 2 pts** No Credit

4. This is only true when the modulus is prime. For modulus $N = pq$, Proposition 3.13 states that there are in fact *four* distinct square roots to any $y \in QR_N$.
5. One way to prove this is as follows: let $\pm u$ be the two square roots of $y \mod p$, and $\pm v$ be the two square roots of $y \mod q$. Proposition 3.13 states that $\textsf{crt}(\pm u, \pm v)$ are the four square roots of $y \mod N$. Our goal is now to show that exactly one of those square roots is in QR_N. To do this, we use Claim 3.1 and then invoke Euler's Criterion (which implies that exactly one of $\pm u$ is in QR_p, and similar for $\pm v$).

CMSC 28400 **Quiz 4 (3 questions)** October 20, 2026

Name: Clarisse Cheung **CNET ID:** clarissec @uchicago.edu

Please write neatly, not too small, and not too faint.
For each problem, you can use the results of prior problem(s) without justification, even if you didn't solve them.

Question 1: Let p be a prime such that $p \equiv 3 \mod 4$. Prove that $-1 \notin QR_p$.

Quadratic residue: $x^2 = y \mod m$

QR of mod 4:

x	x^2	mod 4
1	1	1
2	4	0
3	9	1

QR of mod 4: $\{0, 1\}$. Take $x = 3$ which $\equiv 3 \mod 4$. With $x = 3$, it is still QR = 1. Since -1 is not in QR of mod 4, it cannot be in QR_p.

1

7 49

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Question 2: Let p be a prime such that $p \equiv 3 \mod 4$. Prove that if $y \in QR_p$, then $-y \notin QR_p$.

QR mod 4: $\{0, 1\}$ as shown in QR_p. Any $-y$ can be shown as k = QR value $- y$ → no negatives.

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Question 3: Let $p \neq q$ be distinct primes such that $p, q \equiv 3 \mod 4$ and let $N = pq$. Prove that for any $y \in QR_N$, *exactly one* square root of y is itself a quadratic residue.

must prove ∃ at least 1 and that there cannot be more than 1

Roots when $p \equiv 3 \mod 4$ → $y^{\frac{p+1}{4}}$

∃ 1:

Since $p \equiv 3 \mod 4$, prop 3.12 allows us to calculate any roots: $y^{\frac{p+1}{4}}$ 4

Also because $p \equiv 3 \mod 4$, when p is added by 1, the term $\frac{p+1}{4}$ becomes

↳ break down QR_N into p and q

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